

INDUSTRY 4.0:

Digitization & Smart MRO (Maintenance, Repairing & Overhauling) in Power Systems for Sustainable Energy

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Why Digitization is a Requirement?

Traditional power system audits are manual and reactive, providing limited asset visibility. Industry 4.0 requires audits to evolve into continuous, data-driven assessments based on physical HT/LT asset condition, enabling early risk detection, real-time visibility, predictive insights, and actionable maintenance decisions.

The impact of failure in the Electrical distribution can be tremendous

56%

Of all electrical fires are due to lack of proper maintenance.
Source: Factory Muual Insurance Group

1st

Cause of fire in industrial sites .22% of fires are due to electrical equipment failure
Source: FM global

59%

Of Total Business Interruption losses reported is due to fires & Explosion
Source: Allianz Claims 2015 Dashboard

Why Digitization is a Requirement?

Challenges:

- **Busbar Overheating**

  High current, vibration, loose joints, and corrosion cause busbar overheating, insulation damage, and fire risks—often undetected until failure.

- **Dependence on Manual & Periodic Inspections**

  Thermographic inspections are manual, infrequent, require panel opening, and provide limited, error-prone snapshot data.

- **Lack of Continuous Asset Visibility**

  Undetected busbar overheating can lead to major failures, causing extended shutdowns and production losses.

- **Hidden Moisture Ingress in Control Panels**

  High panel humidity causes condensation, corrosion, and insulation degradation, often unnoticed during manual inspections.

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Main causes of failures and Customer pain points

- Loose joints due to vibrations, unusual operating shocks
- Power cable loose connections due to severe short circuits and aged clamping arrangement
- Mechanical damage of sliding power contacts during equipment handling outside the panels
- Contacts resistance increase caused by oxidation or corrosion due to unfavourable environmental conditions
- Long maintenance intervals due to equipment utilization in continuous process plants.
- Abrasion of contact surface of sliding power contacts due to excess of racking operations above the prescribed limits



Solutions

- Physical Audit
- Continuous Monitoring
- Predictive Analytics
- Prescriptive Action
- Overhauling, Retrofitting & Spares

Smart Busbar Thermal Monitoring

Deploy wireless temperature sensors on busbar joints for continuous monitoring and early alerts, preventing overheating, failures, and fire risks.

Automated, Continuous Wireless Monitoring

Permanently installed sensors stream real-time temperature data to HMI/SCADA dashboards, enabling continuous audits without manual inspections.

Predictive, Integrated Monitoring

Permanently installed sensors stream real-time data to HMI/SCADA, enabling continuous audits without manual inspections.

Continuous Humidity Monitoring

Wireless humidity and temperature sensors continuously monitor panels, trigger early alerts, and prevent corrosion, insulation failure, and breakdowns.

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End-to-End Digitized Power System Solutions

From Intelligent Monitoring to Complete Asset Lifecycle Management which includes overhauling as well as support with BMC manufactured MV Components as OEM

Digitization is most effective when it is supported by physical domain expertise. At **BMC**, Industry 4.0 digitization is delivered as a complete, end-to-end solution—combining sensor-based monitoring, engineering insights, and on-ground execution across the entire lifecycle of electrical panels and their components.

Audit & Digitization

Pre-Digitization Condition Audit

Physical inspection and health assessment of HT/LT assets to identify baseline risks before digitization.

Continuous Condition Assessment

Real-time monitoring of critical components using thermal and humidity sensors to track asset health continuously.

Predictive Fault Detection & Insights

Early identification of physical degradation and thermal abnormalities through audits and digital analytics.

Actionable Lifecycle Solutions (Post-Detection)

Timely servicing, maintenance, repair, spares support, or replacement to ensure complete asset lifecycle management

Predictive Insights & Actionable Maintenance

Targeted Maintenance Planning

Maintenance planning is driven by asset health assessment and predictive analytics, not assumptions or fixed schedules.

Corrective & Lifecycle Actions

Identified abnormalities trigger servicing, repair, retrofitting & spares support, or replacement—ensuring complete lifecycle management.

Reduced Manual Dependency with Faster Decisions

Automated data collection and real-time insights minimize manual intervention while enabling quicker, more confident engineering decisions.

Optimized Asset Performance & Reliability

Condition-driven actions improve system reliability, extend equipment life, and enhance overall energy efficiency.

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What is WB TMS

A Wireless Busbar Temperature Monitoring System is an intelligent and proactive solution for critical electrical systems that continuously monitors temperature at busbar joints and wirelessly transmits real-time data to a controller. By enabling early detection of abnormal heating, it enhances electrical safety, prevents overheating-related failures, minimizes unplanned downtime, and offers a non-intrusive, easy-to-install approach to reliable power system monitoring.

Features

- **Wireless sensors**
- **Max 125°C measurement**
- **Up-to 3 metres range**
- **Battery life lasting upto 5 years**
- **Silicon Strap withstands Temperatures upto 350°C**
- **Integration with SCADA systems**
- **Alerting System (2 relays for set point and break point)**
- **Control panel Temperature & Humidity**
- **Plug and Play installation (15-30 mins)**
- **Compatible with 0.4kV to 66kV systems**







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Humidity Sensors for Control Panel

Features:



- Humidity Monitoring
- Continuous Monitoring
- Wireless Data Transmission



- Sensor capabilities
- Ranges (0-99%)
- Accuracy (±1%)

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How it Works

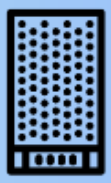


Wireless Communication



Receiver
BBC969-485-R

SMPS (24V, 3Amps)



RS485-RTU



Ethernet/RJ45

Modbus TCP/IP

Scada Control Room



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A Handshake of Digital Software & Electrical Manufacturing Expertise



Comprehensive Monitoring
Covers both temperature and humidity.



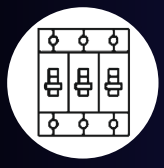
Enhanced Safety
Continuous monitoring ensures system integrity.



User-Friendly Interfaces
Optional controllers with or without displays for versatile monitoring needs.



Technology & Manufacturing Expertise
Four Decades of Power System Excellence



Switchgear Manufacturing Solutions
Vacuum Circuit Breakers & Contactors



Critical Servicing & AMC
Reliable After-Sales & Lifecycle Support

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A Handshake of Digital Software & Electrical Manufacturing Expertise



Cost-Effective Solutions
Reduces manual monitoring costs



Long-Term Reliability
Proven technology with installations operational since 1986



Originality
All our products are patented & proudly designed and manufactured in India.



OEM Manufacturing Capabilities
Instrument Transformers – CT & PT



Cast Resin Component Solutions
Advanced Components & Sub-Assemblies



HT VCB Retrofits & Customization
Tailor-Made CT/PT & Retrofit Solutions





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